

Resumen de modelos

Sin ansatz

Para cada modelo se muestran P^{abcd} y $P_{ab} \equiv M_{ab}$ en forma covariante.

R

$$P^{abcd} = -\frac{g^{a d} g^{b c}}{2} + \frac{g^{a c} g^{b d}}{2}$$

$$\begin{aligned} P_{ab} &\equiv M_{ab} \\ &= 2 R_{a d_0 b}{}^{d_0} \end{aligned}$$

R_plus_alpha_times_R_squared

$$P^{abcd} = -g^{a d} g^{b c} R_{d_0 d_1}{}^{d_0 d_1} \alpha - \frac{g^{a d} g^{b c}}{2} + \frac{g^{a c} g^{b d}}{2} + g^{a c} g^{b d} R_{d_0 d_1}{}^{d_0 d_1} \alpha$$

$$\begin{aligned} P_{ab} &\equiv M_{ab} \\ &= 2 R_{a d_0 b}{}^{d_0} + 4 R_{a d_0 b}{}^{d_0} R_{d_1 d_2}{}^{d_1 d_2} \alpha \end{aligned}$$

R_plus_alpha_times_RicciSq

$$P^{abcd} = -\frac{g^{a d} g^{b c}}{2} - \frac{g^{a d} R_{d_0}{}^b{}_{c d_0} \alpha}{2} - \frac{R_{d_0}{}^a{}_{d_0} g^{b c} \alpha}{2} + \frac{g^{a c} g^{b d}}{2} + \frac{g^{a c} R_{d_0}{}^b{}_{d_0} \alpha}{2} + \frac{R_{d_0}{}^a{}_{d_0} g^{b d} \alpha}{2}$$

$$\begin{aligned} P_{ab} &\equiv M_{ab} \\ &= 2 \left(R_{a d_0 b}{}^{d_0} + \alpha \left(R_{a d_0}{}^{d_0 d_2} R_{b d_1}{}^{d_1 d_2} + R_{a}{}^{d_1}{}_{b}{}^{d_2} R_{d_0 d_1}{}^{d_0 d_2} \right) \right) \end{aligned}$$

R_plus_alpha_times_RiemannSq

$$P^{abcd} = -\frac{g^{a d} g^{b c}}{2} + \frac{g^{a c} g^{b d}}{2} - \frac{2 R^{a d b c} \alpha}{3} + \frac{2 R^{a c b d} \alpha}{3} + \frac{4 R^{a b c d} \alpha}{3}$$

$$\begin{aligned} P_{ab} &\equiv M_{ab} \\ &= 2 R_{a d_0 b}{}^{d_0} + 4 R_{a}{}^{d_0 d_1 d_2} R_{b d_0 d_1 d_2} \alpha \end{aligned}$$

R_plus_alpha_times_RicciUU

$$P^{abcd} = -\frac{g^{a d} g^{b c}}{2} + \frac{g^{a c} g^{b d}}{2} - \frac{g^{a d} u^b u^c \alpha}{4} - \frac{u^a g^{b c} u^d \alpha}{4} + \frac{g^{a c} u^b u^d \alpha}{4} + \frac{u^a g^{b d} u^c \alpha}{4}$$

$$\begin{aligned} P_{ab} &\equiv M_{ab} \\ &= -R_{a d_0}{}^{d_0 d_1} u_b u_{d_1} \alpha - u_a R_{b d_0}{}^{d_0 d_1} u_{d_1} \alpha + 2 R_{a d_0 b}{}^{d_0} + R_{a}{}^{d_0}{}_{b}{}^{d_1} u_{d_0} u_{d_1} \alpha \end{aligned}$$

R_plus_alpha_times_RX

$$P^{abcd} = -\frac{g^{a d} g^{b c} u_{d_0} u^{d_0} \alpha}{2} - \frac{g^{a d} g^{b c}}{2} + \frac{g^{a c} g^{b d} u_{d_0} u^{d_0} \alpha}{2} + \frac{g^{a c} g^{b d}}{2}$$

$$\begin{aligned} P_{ab} &\equiv M_{ab} \\ &= 2 R_{a d_0 b}{}^{d_0} u_{d_1} u^{d_1} \alpha + 2 R_{a d_0 b}{}^{d_0} + u_a u_b R_{d_0 d_1}{}^{d_0 d_1} \alpha \end{aligned}$$

case_07_R_plus_RiemannGrad4

$$P^{abcd} = -\frac{g^{a\,d} g^{b\,c}}{2} + \frac{g^{a\,c} g^{b\,d}}{2}$$

$$\begin{aligned} P_{ab} &\equiv M_{ab} \\ &= 2 R_{a\,d_0\,b}{}^{d_0} \end{aligned}$$

draft4_case_0

$$P^{abcd} = -\frac{g^{a\,d} g^{b\,c}}{2} + \frac{g^{a\,c} g^{b\,d}}{2}$$

$$\begin{aligned} P_{ab} &\equiv M_{ab} \\ &= 2 R_{a\,d_0\,b}{}^{d_0} \end{aligned}$$

draft4_case_1

$$P^{abcd} = -\frac{g^{a\,d} g^{b\,c}}{2} + \frac{g^{a\,c} g^{b\,d}}{2}$$

$$\begin{aligned} P_{ab} &\equiv M_{ab} \\ &= -u_a u_b \alpha_1 + 2 R_{a\,d_0\,b}{}^{d_0} \end{aligned}$$

draft4_case_2

$$\begin{aligned} P^{abcd} = \frac{1}{4} \Big(&-2 g^{a\,d} g^{b\,c} + 2 g^{a\,c} g^{b\,d} \\ &+ \beta_0 \ell^2 \left(-3 g^{a\,d} u^b u^c - 3 u^a g^{b\,c} u^d - 2 g^{a\,c} g^{b\,d} u_{d_0} u^{d_0} + 2 g^{a\,d} g^{b\,c} u_{d_0} u^{d_0} + 3 g^{a\,c} u^b u^d + 3 u^a g^{b\,d} u^c \right) \Big) \end{aligned}$$

$$\begin{aligned} P_{ab} &\equiv M_{ab} \\ &= 2 R_{a\,d_0\,b}{}^{d_0} + \beta_0 \ell^2 \left(-u_a u_b R_{d_0\,d_1}{}^{d_0\,d_1} - 3 R_{a\,d_0}{}^{d_0\,d_1} u_b u_{d_1} - 3 u_a R_{b\,d_0}{}^{d_0\,d_1} u_{d_1} - 2 R_{a\,d_0\,b}{}^{d_0} u_{d_1} u^{d_1} + 3 R_a{}^{d_0\,d_1}{}_{b\,d_0} u_{d_0} u_{d_1} \right) \end{aligned}$$

Con ansatz

Se muestran componentes ya proyectadas y una familia representativa verificada por el solver.

R

Ansatz: draft4_circular.

$$\begin{aligned} \left[P^{abcd} \right]^{0011} &= \frac{1}{2} \\ \left[P^{abcd} \right]^{0022} &= \frac{1}{2 r^2 f(r)} \\ \left[P^{abcd} \right]^{0110} &= -\frac{1}{2} \end{aligned}$$

se muestran 3 de 12 componentes no nulas

$$\begin{aligned} [P_{ab}]_{00} &= \frac{(r f''(r) + f'(r)) f(r)}{r} \\ [P_{ab}]_{11} &= -\frac{(r f''(r) + f'(r))}{r f(r)} \\ [P_{ab}]_{22} &= -2 r f'(r) \end{aligned}$$

Solución de las ecuaciones de campo:

$$f(r) = C_1$$

R_plus_alpha_times_R_squared

Ansatz: draft4_circular.

$$\left[P^{abcd} \right]^{0011} = \frac{(r - 2\alpha (2f'(r) + rf''(r)))}{2r}$$

$$\left[P^{abcd} \right]^{0022} = \frac{(r - 2\alpha (2f'(r) + rf''(r)))}{2r^3 f(r)}$$

$$\left[P^{abcd} \right]^{0110} = \frac{(-\frac{r}{2} + \alpha (2f'(r) + rf''(r)))}{r}$$

se muestran 3 de 12 componentes no nulas

$$[P_{ab}]_{00} = -\frac{1}{r^2} (-r + 2\alpha (2f'(r) + rf''(r))) (rf''(r) + f'(r)) f(r)$$

$$[P_{ab}]_{11} = \frac{(-r + 2\alpha (2f'(r) + rf''(r))) (rf''(r) + f'(r))}{r^2 f(r)}$$

$$[P_{ab}]_{22} = 2 (-r + 2\alpha (2f'(r) + rf''(r))) f'(r)$$

Solución de las ecuaciones de campo:

$$f(r) = C_1$$

R_plus_alpha_times_RicciSq

Ansatz: draft4_circular.

$$\left[P^{abcd} \right]^{0011} = \frac{(r - \alpha (rf''(r) + f'(r)))}{2r}$$

$$\left[P^{abcd} \right]^{0022} = \frac{(2r - \alpha (3f'(r) + rf''(r)))}{4r^3 f(r)}$$

$$\left[P^{abcd} \right]^{0110} = \frac{(-r + \alpha (rf''(r) + f'(r)))}{2r}$$

se muestran 3 de 12 componentes no nulas

$$[P_{ab}]_{00} = \frac{1}{2r^2} f(r) \left(\alpha \left(-3f'(r)^2 - 2r^2 f''(r)^2 - 3rf'(r) f''(r) \right) + 2rf'(r) + 2f''(r) r^2 \right)$$

$$[P_{ab}]_{11} = \frac{1}{r^2 f(r)} \left(-rf'(r) - r^2 f''(r) + \frac{3\alpha f'(r)^2}{2} + \alpha r^2 f''(r)^2 + \frac{3\alpha rf'(r) f''(r)}{2} \right)$$

$$[P_{ab}]_{22} = (-2r + 3\alpha f'(r) + \alpha rf''(r)) f'(r)$$

Solución de las ecuaciones de campo:

$$f(r) = C_1$$

R_plus_alpha_times_RiemannSq

Ansatz: draft4_circular.

$$\left[P^{abcd} \right]^{0011} = -\alpha f''(r) + \frac{1}{2}$$

$$\left[P^{abcd} \right]^{0022} = \frac{(r - 2\alpha f'(r))}{2r^3 f(r)}$$

$$\left[P^{abcd} \right]^{0110} = \alpha f''(r) - \frac{1}{2}$$

se muestran 3 de 12 componentes no nulas

$$[P_{ab}]_{00} = -\frac{1}{r^2} \left(-r (rf''(r) + f'(r)) + 2\alpha (f'(r)^2 + r^2 f''(r)^2) \right) f(r)$$

$$[P_{ab}]_{11} = \frac{1}{r^2 f(r)} \left(-r (rf''(r) + f'(r)) + 2\alpha (f'(r)^2 + r^2 f''(r)^2) \right)$$

$$[P_{ab}]_{22} = 2 (-r + 2\alpha f'(r)) f'(r)$$

Solución de las ecuaciones de campo:

$$f(r) = C_1$$

R_plus_alpha_times_RicciUU

Ansatz: draft4_circular.

$$\begin{aligned} [P^{abcd}]^{0011} &= \frac{\alpha f(r) \left(\frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \right)^2}{4} + \frac{1}{2} \\ [P^{abcd}]^{0012} &= \frac{\alpha \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)}{4 r^2} \\ [P^{abcd}]^{0021} &= \frac{\alpha \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)}{4 r^2} \end{aligned}$$

se muestran 3 de 20 componentes no nulas

$$\begin{aligned} [P_{ab}]_{00} &= \frac{1}{2 r^3} \left(\alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 f'(r) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f''(r) f(r) \right) + 2 r^2 (r f''(r) + f'(r)) \right) f(r) \\ [P_{ab}]_{11} &= -\frac{1}{2 r^3 f(r)} \left(2 r^2 (r f''(r) + f'(r)) + \alpha \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 f'(r) + 2 \alpha r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 (r f''(r) + f'(r)) f(r) \right) \\ [P_{ab}]_{12} &= -\frac{1}{2 r} \alpha (2 f'(r) + r f''(r)) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \end{aligned}$$

se muestran 3 de 5 componentes no nulas

Solución de las ecuaciones de campo:

$$\alpha = 0$$

$$f(r) = C_1$$

R_plus_alpha_times_RX

Ansatz: draft4_circular.

$$\begin{aligned} [P^{abcd}]^{0011} &= \frac{1}{2 r^2} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right) \\ [P^{abcd}]^{0022} &= \frac{1}{2 r^4 f(r)} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right) \\ [P^{abcd}]^{0110} &= -\frac{1}{2 r^2} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right) \end{aligned}$$

se muestran 3 de 12 componentes no nulas

$$\begin{aligned} [P_{ab}]_{00} &= \frac{1}{r^3} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right) (r f''(r) + f'(r)) f(r) \\ [P_{ab}]_{11} &= -\frac{1}{r^3 f(r)} \left(r^2 (r f''(r) + f'(r)) + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) (r f''(r) + f'(r)) + \alpha r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 (2 f'(r) + r f''(r)) f(r) \right) \\ [P_{ab}]_{12} &= -\frac{1}{r} \alpha (2 f'(r) + r f''(r)) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \end{aligned}$$

se muestran 3 de 5 componentes no nulas

Solución de las ecuaciones de campo:

$$\alpha = 0$$

$$f(r) = C_1$$

case_07_R_plus_RiemannGrad4

Ansatz: draft4_circular.

$$\begin{aligned}\left[P^{abcd}\right]^{0011} &= \frac{1}{2} \\ \left[P^{abcd}\right]^{0022} &= \frac{1}{2r^2 f(r)} \\ \left[P^{abcd}\right]^{0110} &= -\frac{1}{2}\end{aligned}$$

se muestran 3 de 12 componentes no nulas

$$\begin{aligned}[P_{ab}]_{00} &= \frac{(r f''(r) + f'(r)) f(r)}{r} \\ [P_{ab}]_{11} &= -\frac{(r f''(r) + f'(r))}{r f(r)} \\ [P_{ab}]_{22} &= -2r f'(r)\end{aligned}$$

Solución de las ecuaciones de campo:

$$f(r) = C_1$$

draft4_case_0

Ansatz: draft4_circular.

$$\begin{aligned}\left[P^{abcd}\right]^{0011} &= \frac{1}{2} \\ \left[P^{abcd}\right]^{0022} &= \frac{1}{2r^2 f(r)} \\ \left[P^{abcd}\right]^{0110} &= -\frac{1}{2}\end{aligned}$$

se muestran 3 de 12 componentes no nulas

$$\begin{aligned}[P_{ab}]_{00} &= \frac{(r f''(r) + f'(r)) f(r)}{r} \\ [P_{ab}]_{11} &= -\frac{(r f''(r) + f'(r))}{r f(r)} \\ [P_{ab}]_{22} &= -2r f'(r)\end{aligned}$$

Solución de las ecuaciones de campo:

$$f(r) = \frac{(r^2 + C_1 \ell^2)}{\ell^2}$$

draft4_case_1

Ansatz: draft4_circular.

$$\begin{aligned}\left[P^{abcd}\right]^{0011} &= \frac{1}{2} \\ \left[P^{abcd}\right]^{0022} &= \frac{1}{2r^2 f(r)} \\ \left[P^{abcd}\right]^{0110} &= -\frac{1}{2}\end{aligned}$$

se muestran 3 de 12 componentes no nulas

$$\begin{aligned}
[P_{ab}]_{00} &= \frac{(r f''(r) + f'(r)) f(r)}{r} \\
[P_{ab}]_{11} &= -\frac{1}{f(r) r} \left(f'(r) + f''(r) r + \alpha_1 f(r) r \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 \right) \\
[P_{ab}]_{12} &= -\alpha_1 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)
\end{aligned}$$

se muestran 3 de 5 componentes no nulas

Solución de las ecuaciones de campo:

$$f(r) = -\frac{(-r^2 + \ell^2 (-C_1 + \alpha_1 q^2 \log(r)))}{\ell^2}$$

draft4_case_2

Ansatz: draft4_circular.

$$\begin{aligned}
[P^{abcd}]^{0011} &= \frac{\beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2}{2 r^2} - \frac{\beta_0 f(r) \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2}{4} - \frac{1}{2} \\
[P^{abcd}]^{0012} &= -\frac{1}{4 r^2} 3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \\
[P^{abcd}]^{0021} &= -\frac{1}{4 r^2} 3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)
\end{aligned}$$

se muestran 3 de 20 componentes no nulas

$$\begin{aligned}
[P_{ab}]_{00} &= \frac{1}{2 r^3} \left(2 r^2 (r f''(r) + f'(r)) + \beta_0 \ell^2 \left(-2 \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) (r f''(r) + f'(r)) + 3 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 f'(r) + 3 r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right) \\
[P_{ab}]_{11} &= \frac{1}{2 r^3 f(r)} \left(-2 f'(r) r^2 - 2 f''(r) r^3 + \beta_0 \ell^2 \left(-f'(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + 2 r f''(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 - 2 f(r) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 \right) \right) \\
[P_{ab}]_{12} &= -\frac{1}{2 r} \beta_0 \ell^2 (2 f'(r) + r f''(r))
\end{aligned}$$

se muestran 3 de 5

Solución de las ecuaciones de campo:

$$f(r) = \frac{r^2 (r^2 + C_1 \ell^2)}{\ell^2 (r^2 + \beta_0 \ell^2 q^2)}$$